

Department of Physics, Princeton University

**Graduate Preliminary Examination
Part I**

Thursday, May 11, 2023
9:00 am - Noon

Answer TWO out of the THREE questions in Section A (Mechanics) and TWO out of the THREE questions in Section B (Electricity and Magnetism).

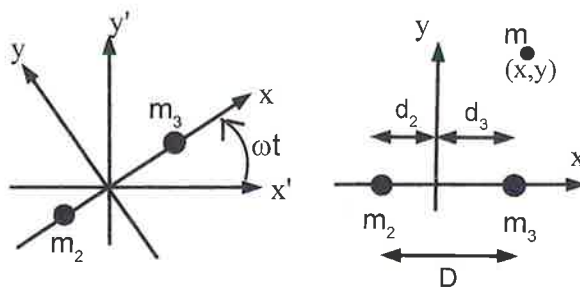
Work each problem in a separate examination booklet.

Be sure to label each booklet with your name, the section name, and the problem number.

Section A. Mechanics

1. Lagrange points in a 3-body problem

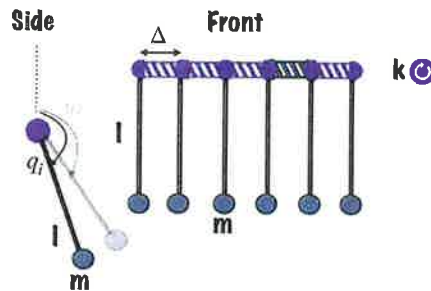
Consider two celestial objects of masses m_2 and m_3 which, owing to their mutual gravitational attraction, rotate around each other without modifying their separation, D . Consider also an asteroid with a mass m which is sufficiently weak with respect to the mass of the two objects that it does not influence their motion. Assume that the mass m moves in the same plane as m_2, m_3 .



On the left, the $x'y'$ frame is the inertial frame where m_2, m_3 orbit each other with fixed separation D . The frame xy is co-rotating with the two masses. On the right, only the rotating frame xy is shown. In this frame m_2, m_3 are static and m orbits the two masses.

- Determine the frequency ω of a rotating frame of reference where the two massive objects are at rest, and determine the distances of the masses d_3 and d_2 from that frame origin. See the figure. Express your answer in terms of D and the masses m_2, m_3 .
- Write an expression for the Hamiltonian $\mathcal{H}(x, y, p_x, p_y)$ of the asteroid of mass m at position (x, y) in this co-rotating frame of reference, where p_x and p_y are the components of the asteroid momentum in that frame. Let $V(x, y)$ be the gravitational potential at the position (x, y) of the asteroid.
- Use Hamilton's equations of motion to get equations determining the equilibrium positions (x, y) of the asteroid (these are known as the Lagrange points of the 3-body problem). Express your result in terms of ω and the derivatives of the gravitational potential $V(x, y)$.
- If you got this far, you should be able to write the equilibrium points as extrema of an effective potential V_{eff} . What is V_{eff} in terms of $V(x, y)$ and ω ? Rewriting this V_{eff} in terms of radii and finding the extrema allows one to find the equilibrium points, but we stop here since the problem is basically solved.

2. Coupled pendulums and Sine-Gordon equation

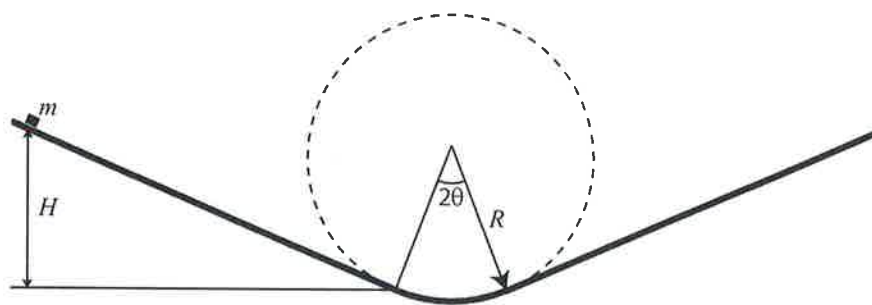


Consider a chain of N identical pendulums of length l and mass m , all suspended from a horizontal axis in a gravitational field g . Each pendulum is separated from the next by a distance Δ . In addition to gravity, the pendulums are subject to a harmonic torque between nearest neighbors $k(q_i - q_{i-1})$, where q_i is the angle between the vertical and the direction of the pendulum numbered i (see the figure), and k is the restoring torque constant.

- Write down the Lagrangian $L(q, \dot{q})$ of the system.
- We now consider the continuous system limit, such that $N \rightarrow \infty$, $m \rightarrow 0$, $\Delta \rightarrow 0$, with the mass density $\mu = m/\Delta$ a finite constant. In this continuum limit, we also let $k\Delta^2/m \equiv \lambda$ be a finite constant. Starting from the Lagrangian in (a), find the Lagrangian density $\mathcal{L}$ of the system in the continuous description.
- Applying the Euler-Lagrange equation for the Lagrangian density, derive the non-linear wave equation (known as the Sine-Gordon equation) satisfied by the system.

3. Inclined track

A small block of mass m is sliding without friction on a track consisting of two straight surfaces inclined at angles $\pm\theta$ and a circular arc of radius R , as shown in the figure. The block starts at a height H above the transition point from straight to circular path. Find the period of oscillations, i.e, the time it takes for the block to return to its starting point. You can assume that angle θ is small.



Section B. Electricity and Magnetism

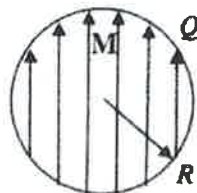
1. Waves in a dielectric medium

Consider plane waves of frequency ω propagating along the $\hat{z}$ direction in an electrically neutral dielectric medium with N molecules per unit volume. Assume each molecule is modelled as an electron of charge q and mass m bound to a fixed ion by a 3D harmonic potential with spring constant κ . There is a large static magnetic field $\mathbf{B} = B\hat{z}$ pointing along the direction of the propagation of the wave. We wish to calculate the index of refraction for the two possible circular polarizations of the wave. In this problem, we will ignore the magnetic field of the wave (convince yourself that this is okay if we assume the electrons are moving non-relativistically). The electric field of the wave can be written as

$$\mathbf{E} = \text{Re} \left(\epsilon E_0 e^{ikz - i\omega t} \right), \quad \epsilon = \hat{x} \pm i\hat{y},$$

where the $+$ sign is for right-handed polarized and the $-$ sign is for left-handed polarized.

- (a) Write down the equation of motion for an electron in one of the molecules in the medium and solve it for the amplitude of the electron's oscillation in the presence of the wave.
- (b) Derive the dispersion relation for the waves in the medium $k(\omega)$, where k is the wavenumber.
- (c) What is the index of refraction of the wave in the medium for each of the two polarizations?

2. A magnetized sphere

Consider a sphere of radius R made from a magnetic material with a permanent uniform magnetization $\mathbf{M}$. The magnetic material is also conducting (e.g. iron) and the sphere carries a total electric charge Q .

- (a) Find the electric field everywhere in space.
- (b) Find the magnetic field outside the sphere.
- (c) Calculate the total angular momentum stored in the electromagnetic fields.

3. Current between conductors in a liquid

- (a) Consider two perfect conductors of arbitrary shape immersed in a large tank of weakly conducting liquid with electrical conductivity σ . Show that the electrical resistance between conductors can be related to their capacitance. For definiteness, assume that the two conductors are held at potentials $+V$ and $-V$ and the walls of the large tank containing the liquid are grounded.
- (b) Consider two conducting spheres with equal radius a immersed in the liquid. The centers of the spheres are separated by a distance $b \gg a$. Find the electrical resistance between the spheres.

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**Graduate Preliminary Examination
Part II**

Friday, May 12, 2023
9:00 am - Noon

Answer TWO out of the THREE questions in Section A (Quantum Mechanics) and TWO out of the THREE questions in Section B (Thermodynamics and Statistical Mechanics).

Work each problem in a separate booklet. Be sure to label each booklet with your name, the section name, and the problem number.

Section A. Quantum Mechanics

1. Perturbed 2d harmonic oscillator

A particle of mass m moves in two dimensions under the influence of a isotropic harmonic potential perturbed by a quadratic anisotropic term. The Hamiltonian is given by

$$H = \frac{p_x^2 + p_y^2}{2m} + \frac{m\omega^2}{2}(x^2 + y^2) + \alpha m\omega^2 xy$$

- (a) Consider first the case $\alpha = 0$. What are the energy levels and their degeneracies?
- (b) Now suppose $|\alpha| \ll 1$. Use first order perturbation theory to find the corrections to the energies of the first two unperturbed energy levels.
- (c) Find the exact energy spectrum of H for finite α , with $0 \leq |\alpha| < 1$. (*Hint:* make a coordinate transformation such that the two degrees of freedom decouple). Check that your result agrees with perturbation theory to first order in α .

2. Rabi oscillations in a two-level system

Consider a two-state system with unperturbed Hamiltonian $\hat{H}_0$ (time independent). We then perturb it by a time dependent potential, so that the Hamiltonian is

$$\hat{H}(t) = \hat{H}_0 + \hat{V}(t)$$

The eigenstates of $\hat{H}_0$ are $|1\rangle$ and $|2\rangle$ with energies E_1, E_2 . Assume $E_1 < E_2$, and denote $\omega_{21} = (E_2 - E_1)/\hbar$. We will take the perturbation to have the form

$$\hat{V}(t) = \gamma e^{i\omega t} |1\rangle\langle 2| + \gamma e^{-i\omega t} |2\rangle\langle 1|$$

where γ is a real constant, and ω a given frequency.

Suppose the system is in the state $|1\rangle$ at time $t = 0$. At later time, we write

$$|\psi(t)\rangle = c_1(t) e^{-\frac{i}{\hbar} E_1 t} |1\rangle + c_2(t) e^{-\frac{i}{\hbar} E_2 t} |2\rangle$$

with $c_1(0) = 1, c_2(0) = 0$.

- Write down the coupled differential equation for $c_1(t), c_2(t)$ that follows from the Schrodinger equation.
- The differential equation can be solved exactly for arbitrary ω . However, to simplify the algebra in this problem we focus on the resonant case $\omega = \omega_{21}$. Explicitly solve the differential equation for $c_1(t), c_2(t)$ in this special case, with the initial conditions stated above.
- What are the probabilities $P_1(t), P_2(t)$ that the system is found in the states $|1\rangle, |2\rangle$ respectively? Sketch a plot of these probabilities as a function of time. At what time is the system first found with certainty in the state $|2\rangle$?
- Suppose γ is small. Use first order time dependent perturbation theory to find the probability $P_2(t)$ that the system is found in the state $|2\rangle$. Work this out for general ω , then take a limit $\omega \rightarrow \omega_{12}$ of the end result and compare with the exact result obtained above. Under what condition is the perturbation theory result for $\omega = \omega_{21}$ valid?

3. Bound states in double delta function potential

A particle with mass m moves in one dimension in the presence of the potential

$$V(x) = -\frac{\hbar^2 \beta}{m} (\delta(x+a) + \delta(x-a))$$

where β is a positive constant with dimension of inverse length. As you will show below, if the dimensionless quantity βa is large enough, this potential has two bound states: the ground state with energy E_0 and one excited state with energy E_1 .

- (a) Write down a suitable ansatz for the ground state wavefunction, and use the appropriate matching conditions at $x = \pm a$ to derive a transcendental equation that determines the ground state energy E_0 . Show that this equation always has a solution.
- (b) Repeat the above for the case of the excited state wavefunction, and derive the equation that determines the energy E_1 of the excited state. Show that this equation only has a solution above a certain critical value of βa . Find this critical value.
- (c) In the limit $\beta a \gg 1$, show that the two energy levels are exponentially close

$$E_1 - E_0 \approx C (e^{-2\beta a} + O(e^{-4\beta a}))$$

and find the constant C explicitly.

Section B. Statistical Mechanics and Thermodynamics

1. Schottky defect model

Consider a system of $M \gg 1$ independent defects in a solid. At each such defect there are two energy levels, one at energy zero and one at energy $\Delta > 0$. Each of these energy levels may be occupied by a particle or be empty, and there are no interactions between the particles. Thus each defect may contain 0, 1 or 2 particles. The M defects are occupied by N particles where $N = M$. The system is at equilibrium at temperature T .

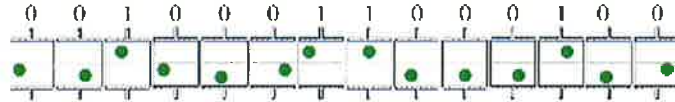
- (a) What is the chemical potential of the particles occupying the defects?
- (b) What is the average energy of the particles occupying the defects?

2. Bernoulli's equation for ideal gas

Bernoulli's equation is usually discussed in relation to incompressible liquid flow, but the same principle of energy conservation along a streamline can be applied to the flow of an ideal compressible gas.

- (a) Find a relationship between mass density ρ and pressure p along a streamline for an adiabatic flow of an ideal gas with $\gamma = C_p/C_V$.
- (b) The flow along a streamline can be parametrized by velocity v and pressure p . Write a differential equation describing energy conservation along the streamline. Integrate this equation to obtain Bernoulli's equation for compressible flow. You can ignore the effects of gravity.
- (c) Consider the gas with molar mass M escaping from a small hole in a compressed gas bottle. Inside the bottle the pressure and temperature are p_1 and T_1 . Outside the bottle the atmospheric pressure is p_2 . The area of the hole is much smaller than the cross-sectional area of the bottle. Find the velocity of the escaping gas.

3. Work from Information



It is possible for information to do work, an example of this is Szilard's machine, sketched in the figure. This version, due to Bennett, is a digital tape where the information is stored in the position of gas molecules. We say the tape is made up of little cells with single gas molecules either in the bottom of the cell, which we call 0, or in the top of the cell which we call 1. Imagine there is a thin dividing membrane between the top and bottom keeping the molecule from moving from top to bottom. Let us keep this molecular tape in thermal contact with a heat bath so that the molecules are always at constant temperature T .

Bennett proposed putting pistons on both the top and bottom of each cell. Say the molecule is on the bottom (i.e., a 0). If so, we lower the top piston to isolate the molecule on the bottom half, this doesn't cost any energy. Then we remove the dividing membrane and let the thermal motion of the molecule slowly push the piston up, drawing heat from the heat bath. Similarly for the molecule on the top of the cell (a 1).

- Calculate the energy per cell E_0 extracted from the heat bath in "erasing" a bit of information, so that we now do not know if the cell contains a 1 or 0.
- Bennett's scheme solves the problem of Maxwell's Demon. Imagine a container with a wall down the middle in contact with a heat bath at temperature T . Each side has volume $V/2$. On each side of the wall are N identical balls, except $N/2$ of them are red, and $N/2$ of them are blue. The Demon slides a door open and shut quickly if a red ball approaches from the left side and thus enters the right side, or if a blue ball approaches from the right side and thus enters the left side. Eventually the container will now have N red balls on the right half, and N blue balls on the left half, yet it would appear the Demon did no work. State the problem with this and use the result of (a) to resolve the problem.

